

Error Log Report

QUESTION 1.

Line No.	Error	Error Type	Correction
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6	Missing semicolon after `int marks[5] = {80,90,70,60,50};`	Syntax Error	`int marks[5] = {80,90,70,60,50};`
12	`i <= 5` accesses `marks[5]` (out of bounds)	Runtime/Logical Error	`for(int i=0; i<5; i++)`
21	`printf("Average = %d\n", average);` uses wrong format specifier	Semantic Error	`printf("Average = %.2f\n", average);`
32	`scores[5] = 60;` exceeds array size	Runtime Error	Remove statement or increase array size
45	`strcpy(subject,"Programming");` overflows array `subject[5]`	Runtime Error	Increase size: `char subject[20];`
54	`attendance["one"] = 1;` uses string as index	Syntax/Semantic Error	Use integer index, e.g., `attendance[1] = 1;`
56	`attendance[2.5] = 1;` uses float index	Syntax/Semantic Error	Use integer index, e.g., `attendance[2] = 1;`
65	`i <= 5` accesses `attendance[5]`	Runtime Error	`for(int i=0; i<5; i++)`
75	`scanf("%d", arr[i]);` missing address operator	Runtime Error	`scanf("%d", &arr[i]);`
91	`grade[0] = "A";` assigns string to char	Syntax Error	`grade[0] = 'A';`
94	`printf("%s\n", grade);` string may not be null terminated	Runtime Error	Add `grade[2] = '\0';` before printing
100	`if(prices[2] = 300)` uses assignment instead of comparison	Logical Error	`if(prices[2] == 300)`
109	`inventory = prices;` array assignment not allowed	Syntax Error	Copy using loop or `memcpy()`
118	`printf("%d\n", salary[0]);` wrong format specifier for float	Semantic Error	`printf("%.2f\n", salary[0]);`
124	`city = "Jaipur";` array assignment not allowed	Syntax Error	`strcpy(city, "Jaipur");`
133	`i <= 5` writes to `result[5]`	Runtime Error	`for(int i=0; i<5; i++)`
20	`average = total / 5;` performs integer division	Logical Error	`average = (float)total / 5;`
89	`char grade[5];` not initialized before string print	Runtime Error	Initialize properly: `char grade[5] = "A+";`
108	`inventory[0]` used without valid initialization	Runtime Error	Copy values from `prices` first
123	`city` declared but not initialized before assignment attempt	Semantic Error	Use `strcpy(city, "Jaipur");`
Entire File	Same program repeated multiple times	Syntax Error	Keep only one complete program in the source file

QUESTION 2.

Line No.	Error	Error Type	Correction
15	<code>for(int i=0;i<=10;i++)` accesses `numbers[10]`</code>	Runtime Error (Array Out of Bounds)	<code>for(int i=0;i<10;i++)`</code>
25	<code>sum = numbers[i];` only stores last value instead of adding</code>	Logical Error	<code>sum += numbers[i];`</code>
33	<code>printf("Average = %d\n", average);` wrong format specifier for float</code>	Semantic Error	<code>printf("Average = %.2f\n", average);`</code>
31	<code>average = sum / 10;` performs integer division</code>	Logical Error	<code>average = (float)sum / 10;`</code>
44	<code>if(marks[i] < highest)` finds minimum instead of highest</code>	Logical Error	<code>if(marks[i] > highest)`</code>
60	<code>attendance[5] = 1;` exceeds array size</code>	Runtime Error	Remove statement or increase array size
64	<code>for(int i=0;i<=5;i++)` accesses `attendance[5]`</code>	Runtime Error	<code>for(int i=0;i<5;i++)`</code>
74	<code>strcpy(subject,"ComputerScience");` string larger than array size 10</code>	Runtime Error	Increase array size: <code>char subject[20];`</code>
82	<code>branch = "IT";` array assignment not allowed</code>	Syntax Error	<code>strcpy(branch,"IT");`</code>
96	<code>printf("%s\n", college);` string not null terminated</code>	Runtime Error	Add <code>college[5] = '\0';`</code>
102	<code>fees["two"] = 50000;` string used as array index</code>	Syntax/Semantic Error	Use integer index, e.g., <code>fees[2] = 50000;`</code>
104	<code>fees[3.5] = 60000;` float used as array index</code>	Syntax/Semantic Error	Use integer index, e.g., <code>fees[3] = 60000;`</code>
122	<code>scanf("%d", result[i]);` missing address operator</code>	Runtime Error	<code>scanf("%d", &result[i]);`</code>
139	<code>if(stock[2] = 300)` assignment instead of comparison</code>	Logical Error	<code>if(stock[2] == 300)`</code>
148	<code>backup = stock;` array assignment not allowed</code>	Syntax Error	Copy using loop or <code>memcpy()</code>
157	<code>printf("%d\n", salary[0]);` wrong format specifier for float</code>	Semantic Error	<code>printf("%.2f\n", salary[0]);`</code>
163	<code>for(int i=0;i<=5;i++)` accesses `inventory[5]`</code>	Runtime Error	<code>for(int i=0;i<5;i++)`</code>
177	<code>strcpy(city,"JaipurCity");` string exceeds array size 8</code>	Runtime Error	Increase size: <code>char city[11];`</code>
185	<code>count = inventory;` array assignment not allowed</code>	Syntax Error	Copy elements using loop
191	Missing semicolon after <code>printf("Program Ended\n")`</code>	Syntax Error	<code>printf("Program Ended\n");`</code>

QUESTION 3.

Line No.	Error	Error Type	Correction
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19	`average = total / 5;` performs integer division	Logical Error	`average = (float)total / 5;`
40	`if(marks[i] > lowest)` finds highest instead of lowest	Logical Error	`if(marks[i] < lowest)`
58	`for(int i=0;i<=5;i++)` accesses `attendance[5]`	Runtime Error	`for(int i=0;i<5;i++)`
68	`char subject[8];` too small for `"Programming"`	Runtime Error	`char subject[12];` (or larger)
70	`strcpy(subject,"Programming");` causes buffer overflow	Runtime Error	Increase array size before copying
78	`scanf("%s",&branch);` incorrect argument	Semantic Error	`scanf("%9s", branch);`
91	`fees[5] = 100000;` exceeds array size	Runtime Error	Remove statement or increase array size
103	`if(stock[2] = 30)` uses assignment instead of comparison	Logical Error	`if(stock[2] == 30)`
112	`backup = stock;` array assignment not allowed	Syntax Error	Copy elements using loop or `memcpy()`
121	`printf("%d\n", salary[0]);` wrong format specifier for float	Semantic Error	`printf("%.2f\n", salary[0]);`
129	`scanf("%d", result[i]);` missing address operator	Runtime Error	`scanf("%d", &result[i]);`
145	`city = "Jaipur";` array assignment not allowed	Syntax Error	`strcpy(city, "Jaipur");`
153	`inventory["one"] = 100;` string used as array index	Syntax/Semantic Error	`inventory[1] = 100;`
155	`inventory[2.5] = 200;` float used as array index	Syntax/Semantic Error	`inventory[2] = 200;`
165	`for(int i=0;i<=5;i++)` accesses `inventory[5]`	Runtime Error	`for(int i=0;i<5;i++)`
177	`char college[5];` no space for null terminator	Runtime Error	`char college[6];`
184	`printf("%s\n", college);` string not null terminated	Runtime Error	Add `college[5] = '\0';`
190	`count = inventory;` array assignment not allowed	Syntax Error	Copy values using loop
198	`char course[6];` too small for `"Computer"`	Runtime Error	`char course[9];`
200	`strcpy(course,"Computer");` causes buffer overflow	Runtime Error	Increase array size
208	`for(int i=0;i<=5;i++)` accesses `numbers[5]`	Runtime Error	`for(int i=0;i<5;i++)`
220	Missing semicolon after `return 0`	Syntax Error	`return 0;`
Entire File Program repeated multiple times in same source file Syntax Error Keep only one complete program			

QUESTION 4.

Line No.	Error	Error Type	Error Correction
20	<code>`printf("Average Sales = %d\n", averageSales);`</code>	Semantic Error	Use <code>`%f`</code> for float:
28	<code>`printf("Average Sales = %.2f\n", averageSales);`</code>		
28	<code>`if(sales[i] < highestSale)`</code>	Logical Error	Use <code>`>`</code> to find highest sale: <code>`if(sales[i] > highestSale)`</code>
45	<code>`monthlySales[5] = 250;`</code>	Runtime Error (Array Out of Bounds)	Use valid index (0–4) or increase array size.
50	<code>`for(int i=0;i<=5;i++)`</code>	Runtime Error	Change to <code>`for(int i=0;i<5;i++)`</code>
60	<code>`char product[8];`</code> with <code>`"LaptopComputer"`</code>	Runtime Error (Buffer Overflow)	Increase size: <code>`char product[20];`</code>
70	<code>`category = "Electronics";`</code>	Syntax Error	Use <code>`strcpy(category, "Electronics");`</code>
78	<code>`stock["first"] = 100;`</code>	Syntax Error	Use integer index: <code>`stock[0] = 100;`</code>
80	<code>`stock[2.7] = 200;`</code>	Syntax Error	Use integer index: <code>`stock[2] = 200;`</code>
101	<code>`scanf("%d", orders[i]);`</code>	Runtime Error	Pass address: <code>`scanf("%d", &orders[i]);`</code>
118	<code>`if(revenue[2] = 3000)`</code>	Logical Error	Use comparison operator: <code>`if(revenue[2] == 3000)`</code>
127	<code>`backupRevenue = revenue;`</code>	Syntax Error	Copy elements using loop or <code>`memcpy()`</code>
135	<code>`printf("%d\n", commission[0]);`</code>	Semantic Error	Use <code>`%f`</code> : <code>`printf("%f\n", commission[0]);`</code>
141	<code>`city = "Jaipur";`</code>	Syntax Error	Use <code>`strcpy(city, "Jaipur");`</code>
152	<code>`printf("%s\n", branch);`</code>	Runtime Error	Add null terminator: <code>`branch[4] = '\0';`</code>
160	<code>`for(int i=0;i<=5;i++)`</code>	Runtime Error	Change to <code>`for(int i=0;i<5;i++)`</code>
176	<code>`char department[6];`</code> with <code>`"Marketing"`</code>	Runtime Error (Buffer Overflow)	Increase size: <code>`char department[10];`</code> or larger.
184	<code>`expenses = sales;`</code>	Syntax Error	Copy array elements using loop or <code>`memcpy()`</code>
192	<code>`ratings[-1] = 5;`</code>	Runtime Error (Invalid Index)	Use valid index: <code>`ratings[0] = 5;`</code>
204	<code>`printf("%s\n", feedback[0]);`</code>	Semantic Error	Use <code>`%d`</code> : <code>`printf("%d\n", feedback[0]);`</code>
208	<code>`printf("Report Generated Successfully\n")`</code>	Syntax Error	Add semicolon: <code>`printf("Report Generated Successfully\n");`</code>

QUESTION 5.

Line No.	Error	Error Type	Error Correction
20	<code>`printf("Average = %d\n", average);`</code>	Semantic Error	Use <code>`%f`</code> for float: <code>`printf("Average = %.2f\n", average);`</code>
28	<code>`if(marks[i] < highest)`</code>	Logical Error	Use <code>`>`</code> to find highest mark: <code>`if(marks[i] > highest)`</code>
40	<code>`if(marks[i] > lowest)`</code>	Logical Error	Use <code>`<`</code> to find lowest mark: <code>`if(marks[i] < lowest)`</code>
56	<code>`attendance[10] = 0;`</code>	Runtime Error (Array Out of Bounds)	Valid indices are 0–9. Use a valid index or increase array size.
60	<code>`for(int i=0;i<=10;i++)`</code>	Runtime Error	Change to <code>`for(int i=0;i<10;i++)`</code>
70	<code>`char subject[10];`</code> with <code>`"DataStructures"`</code>	Runtime Error (Buffer Overflow)	Increase size: <code>`char subject[15];`</code> or larger.
78	<code>`department = "InformationTechnology";`</code>	Syntax Error	Use <code>`strcpy(department, "InformationTechnology");`</code>
86	<code>`fees["two"] = 50000;`</code>	Syntax Error	Use integer index: <code>`fees[2] = 50000;`</code>
88	<code>`fees[4.5] = 70000;`</code>	Syntax Error	Use integer index: <code>`fees[4] = 70000;`</code>
109	<code>`scanf("%d", result[i]);`</code>	Runtime Error	Pass address: <code>`scanf("%d", &result[i]);`</code>
126	<code>`if(inventory[2] = 30)`</code>	Logical Error	Use comparison operator: <code>`if(inventory[2] == 30)`</code>
135	<code>`backup = inventory;`</code>	Syntax Error	Copy array elements using a loop or <code>`memcpy()`</code> .
143	<code>`printf("%d\n", salary[0]);`</code>	Semantic Error	Use <code>`%f`</code> : <code>`printf("%f\n", salary[0]);`</code>
149	<code>`city = "Jaipur";`</code>	Syntax Error	Use <code>`strcpy(city, "Jaipur");`</code>
160	<code>`printf("%s\n", branch);`</code>	Runtime Error	Add null terminator: <code>`branch[4] = '\0';`</code>
168	<code>`for(int i=0;i<=5;i++)`</code>	Runtime Error	Change to <code>`for(int i=0;i<5;i++)`</code>
184	<code>`char course[8];`</code> with <code>`"Programming"`</code>	Runtime Error (Buffer Overflow)	Increase size: <code>`char course[12];`</code> or larger.
194	<code>`ratings[-1] = 10;`</code>	Runtime Error (Invalid Index)	Use a valid index such as <code>`ratings[0] = 10;`</code>
200	<code>`printf("%s\n", feedback[0]);`</code>	Semantic Error	Use <code>`%d`</code> : <code>`printf("%d\n", feedback[0]);`</code>
208	<code>`scores = marks;`</code>	Syntax Error	Copy array elements using a loop or <code>`memcpy()`</code> .
220	<code>`printf("%d\n", temp[5]);`</code>	Runtime Error (Array Out of Bounds)	Use a valid index (0–4).